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4441 trading days

From: 2007-03-30

To: 2026-09-02

📅 Earliest 3 trading days:

	S&P 500	EURO STOXX 50	Nikkei 225	FTSE 100
Date				
2007-03-30	1420.859985	4181.029785	17287.650391	6308.000000
2007-04-02	1424.550049	4189.549805	17028.410156	6315.500000
2007-04-03	1437.770020	4246.299805	17244.050781	6366.100098

📅 Latest 3 trading days (most current):

	S&P 500	EURO STOXX 50	Nikkei 225	FTSE 100
Date				
2026-08-28	7711.759766	6485.669922	66405.562500	10824.299805
2026-09-01	7631.470215	6368.979980	66215.343750	10789.299805
2026-09-02	7666.600098	6362.149902	64325.640625	10756.500000

Returns matrix: 4440 days, 4 indices

Descriptive Statistics (Daily % Returns):

	count	mean	std	min	25%	50%	75%	max
S&P 500	4440.0	0.037964	1.293991	-12.765220	-0.430939	0.080073	0.617858	10.423562
EURO STOXX 50	4440.0	0.009455	1.437538	-13.240515	-0.627174	0.049665	0.697948	12.950880
Nikkei 225	4440.0	0.029594	1.559034	-13.234076	-0.714252	0.069444	0.835828	13.234592
FTSE 100	4440.0	0.012020	1.177786	-11.511706	-0.484072	0.058403	0.575108	11.111187

First 3 rows of returns:

	S&P 500	EURO STOXX 50	Nikkei 225	FTSE 100
Date				
2007-04-02	0.259370	0.203571	-1.510926	0.118826
2007-04-03	0.923731	1.345469	1.258407	0.798012
2007-04-04	0.111220	0.365069	1.724993	-0.021992

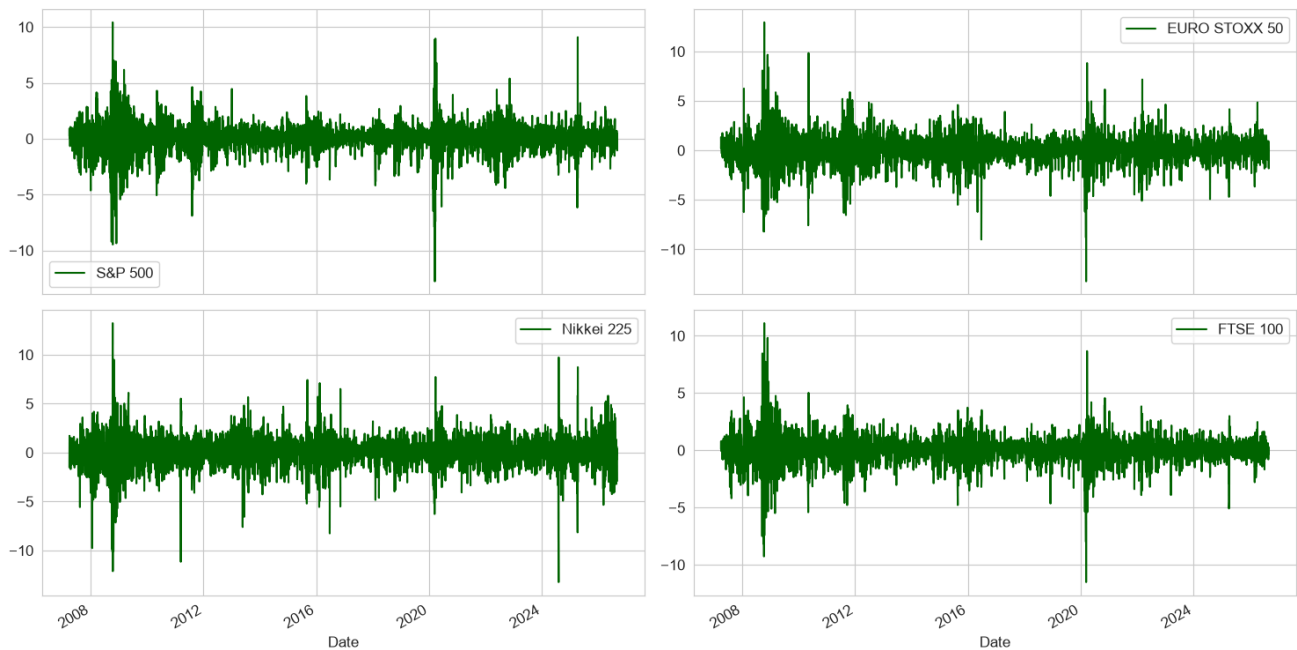
Last 3 rows of returns:

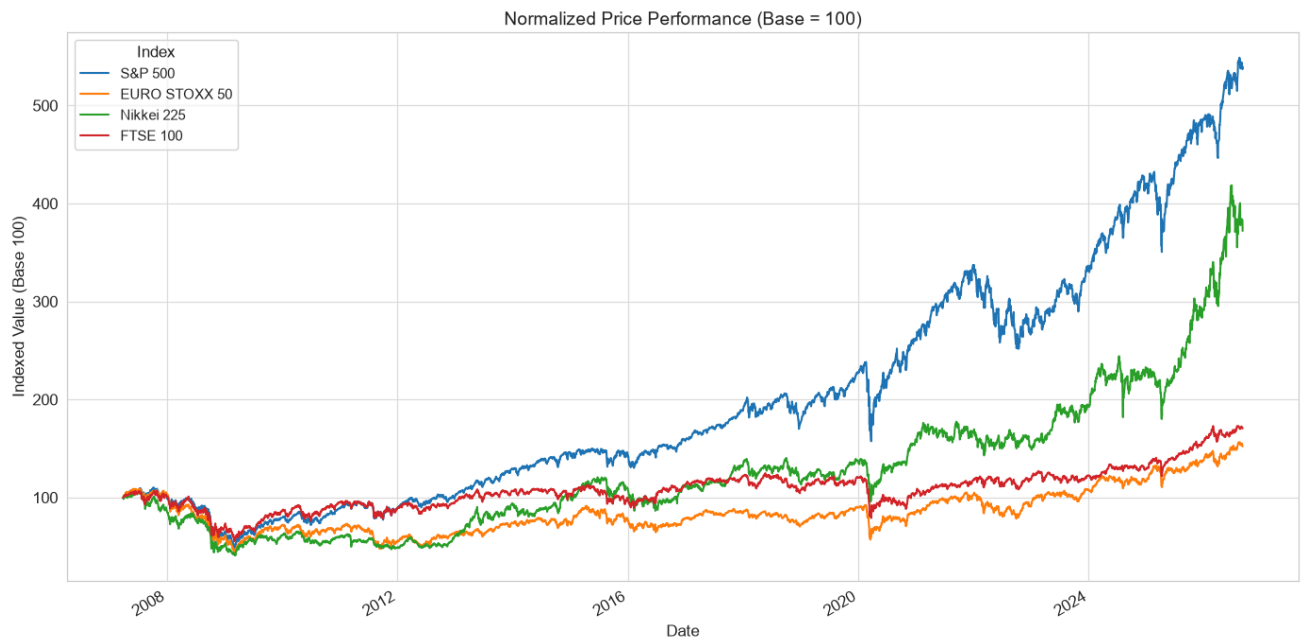
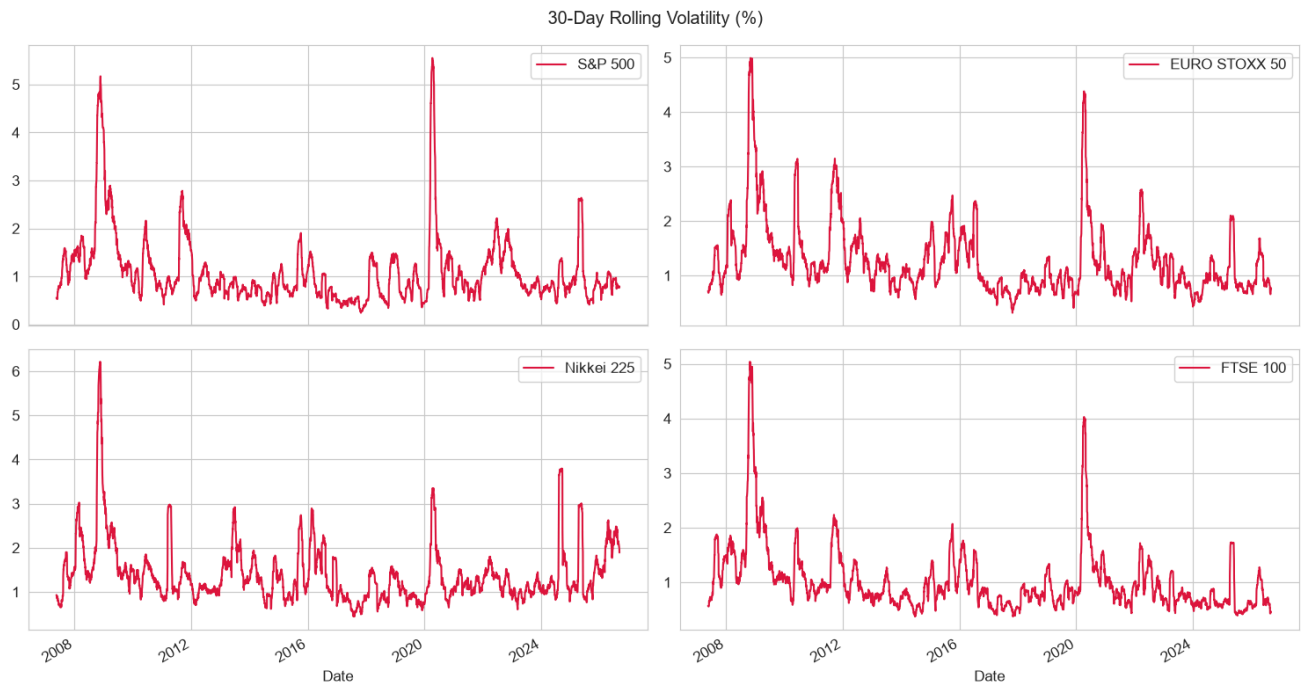
	S&P 500	EURO STOXX 50	Nikkei 225	FTSE 100
Date				
2026-08-28	-0.249055	0.944051	0.412844	0.294214
2026-09-01	-1.046589	-1.815579	-0.286861	-0.323870
2026-09-02	0.459273	-0.107297	-2.895390	-0.304466

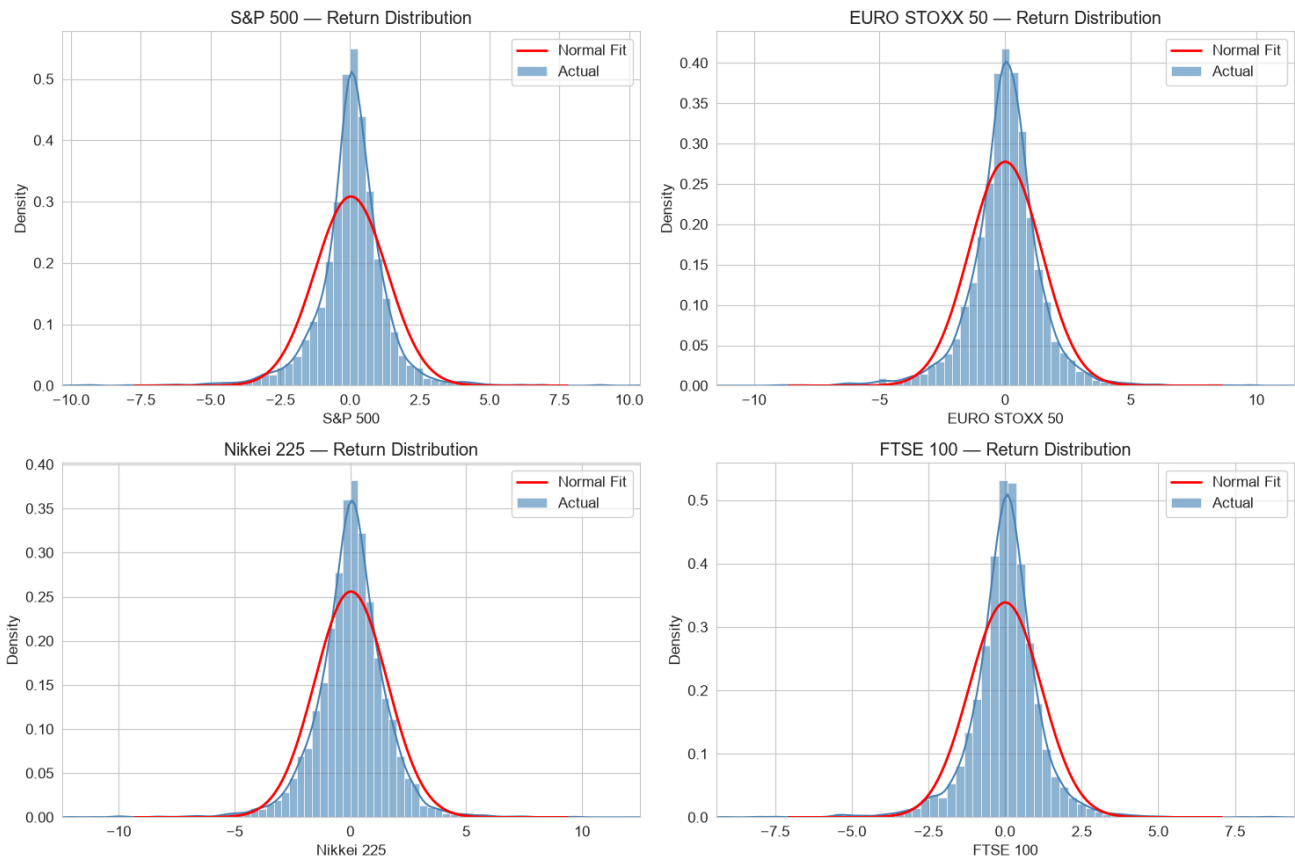
Adjusted Close Prices



Daily Log Returns (%)







=== Distribution Shape Summary ===

(Normal distribution: Excess Kurtosis = 0, Skewness = 0)

	Excess Kurtosis	Skewness	Std Dev (%)
S&P 500	11.169	-0.625	1.294
EURO STOXX 50	8.083	-0.258	1.438
Nikkei 225	8.003	-0.454	1.559
FTSE 100	11.257	-0.301	1.178

Fitting GARCH(1,1) with Normal errors for S&P 500...

S&P 500 fit complete. AIC = 12555.45, BIC = 12581.04

Fitting GARCH(1,1) with Normal errors for EURO STOXX 50...

EURO STOXX 50 fit complete. AIC = 14344.71, BIC = 14370.31

Fitting GARCH(1,1) with Normal errors for Nikkei 225...

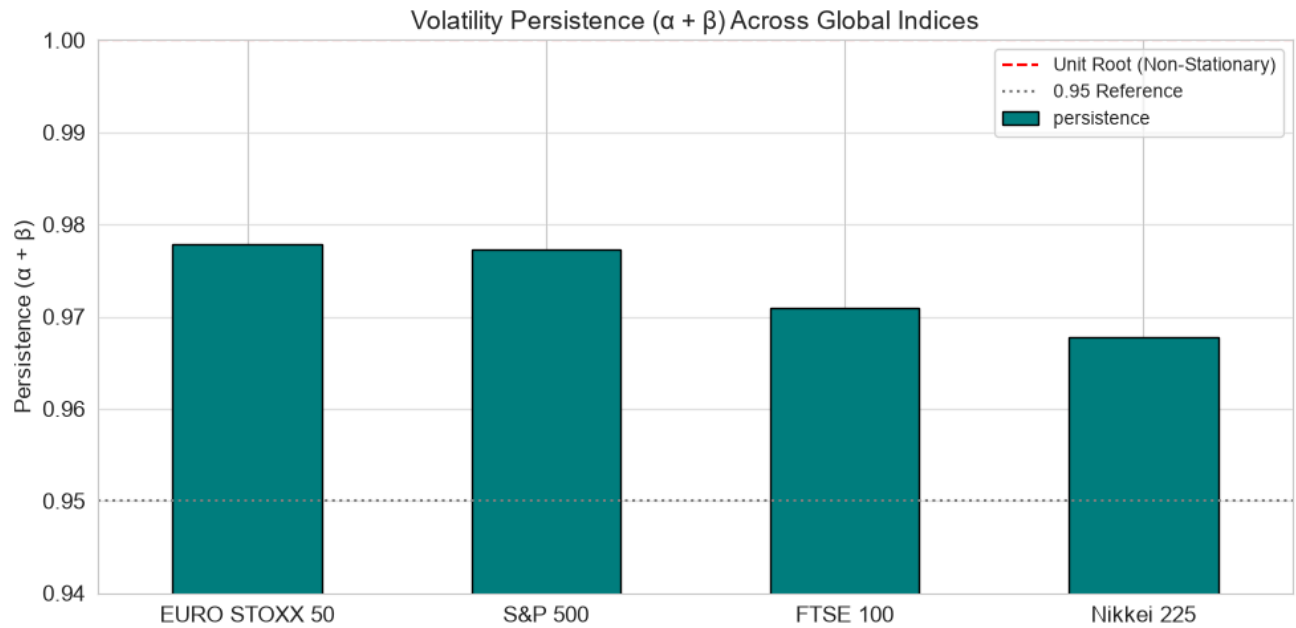
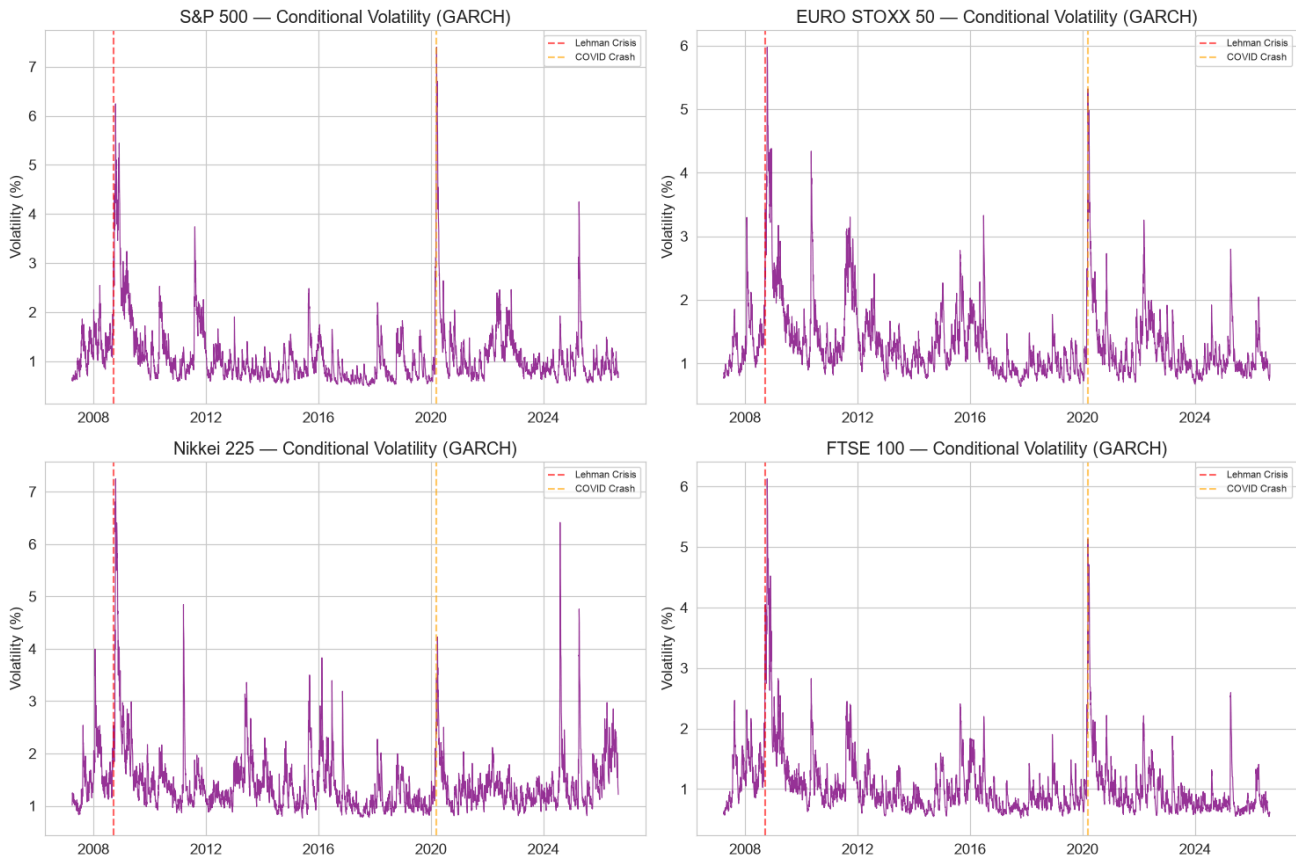
Nikkei 225 fit complete. AIC = 15265.62, BIC = 15291.21

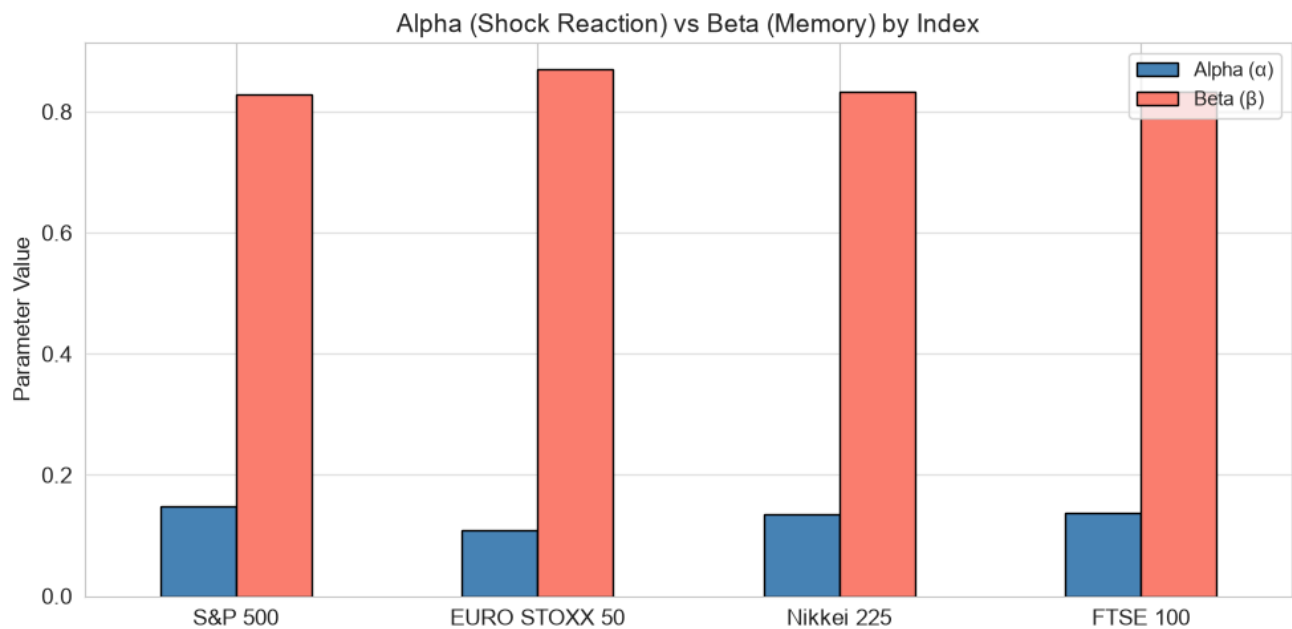
Fitting GARCH(1,1) with Normal errors for FTSE 100...

FTSE 100 fit complete. AIC = 12183.62, BIC = 12209.21

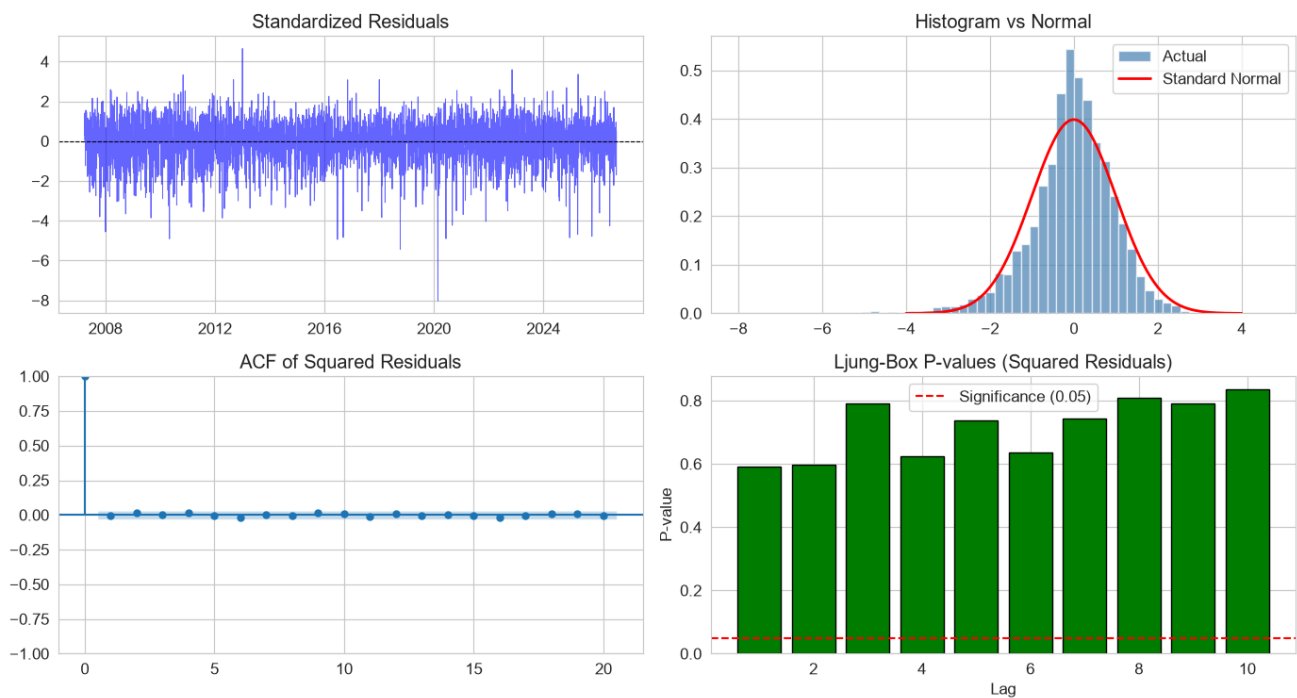
=== GARCH(1,1)-Normal Parameter Estimates ===

	mu	omega	alpha	beta	persistence	AIC	BIC
S&P 500	0.079719	0.037072	0.148216	0.829144	0.977360	12555.448859	12581.042497
EURO STOXX 50	0.053853	0.045430	0.108211	0.869725	0.977936	14344.713752	14370.307391
Nikkei 225	0.071471	0.081006	0.135571	0.832269	0.967840	15265.617812	15291.211450
FTSE 100	0.040745	0.037704	0.138088	0.832907	0.970996	12183.618439	12209.212077





S&P 500 — Model Diagnostics

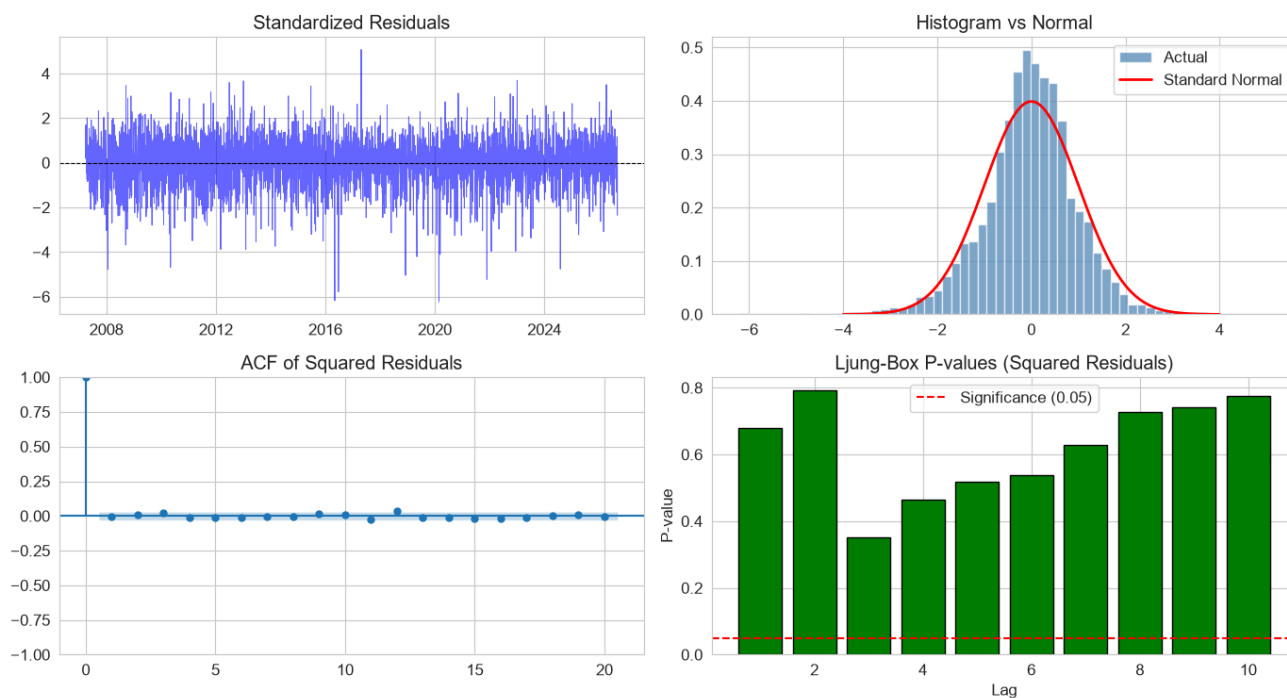


S&P 500 Ljung-Box test (squared residuals, lags 1-10):

P-values: [0.5926 0.5982 0.7911 0.6232 0.739 0.6356 0.744 0.8081 0.7924 0.8355]

All p-values > 0.05. Cannot reject H_0 of no autocorrelation. Model is adequate.

EURO STOXX 50 — Model Diagnostics

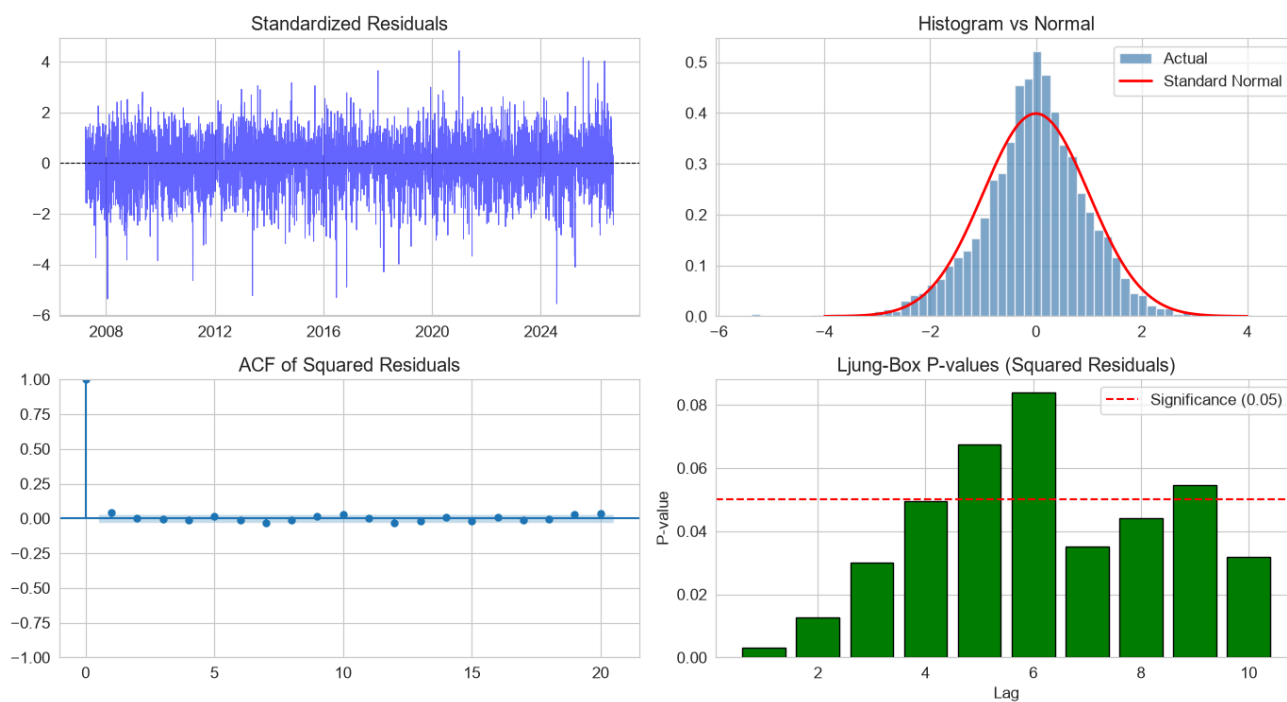


EURO STOXX 50 Ljung-Box test (squared residuals, lags 1-10):

P-values: [0.6798 0.792 0.3513 0.464 0.5171 0.5372 0.6296 0.7284 0.7418 0.7751]

All p-values > 0.05. Cannot reject H_0 of no autocorrelation. Model is adequate.

Nikkei 225 — Model Diagnostics

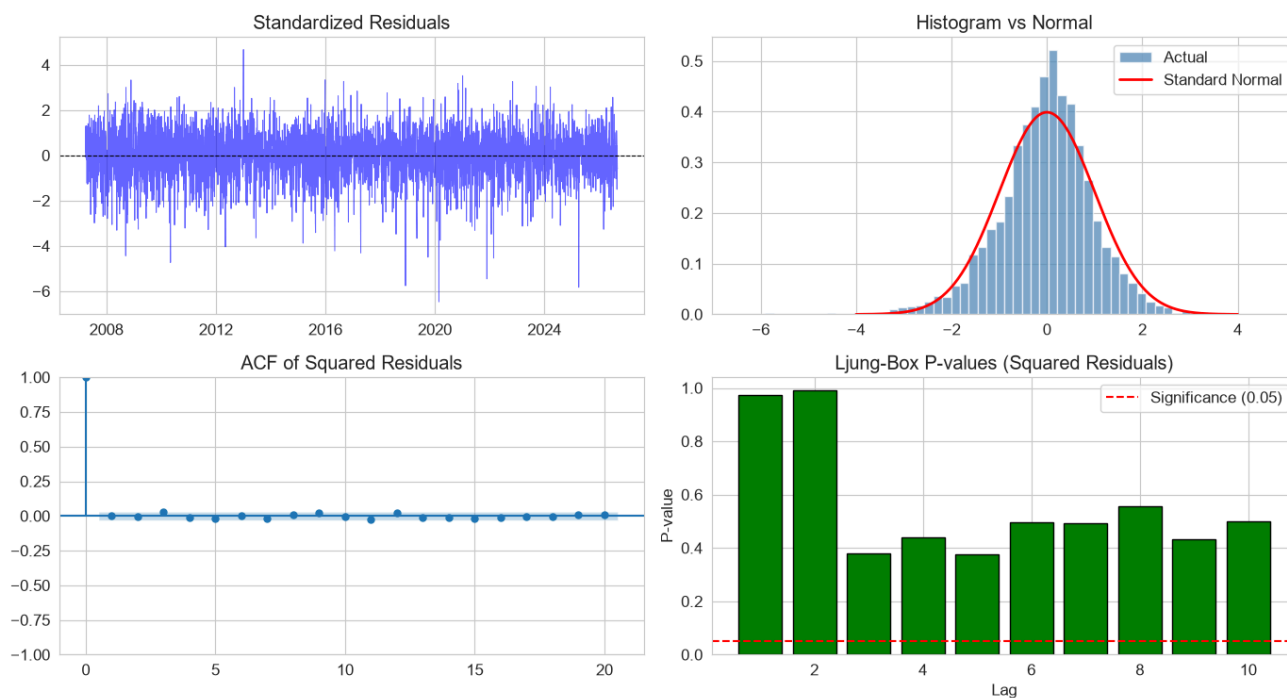


Nikkei 225 Ljung-Box test (squared residuals, lags 1-10):

P-values: [0.0031 0.0126 0.0299 0.0497 0.0676 0.0839 0.0351 0.0441 0.0547 0.0317]

Some p-values < 0.05. Remaining ARCH effects detected. Consider higher-order GARCH.

FTSE 100 — Model Diagnostics



FTSE 100 Ljung-Box test (squared residuals, lags 1-10):

P-values: [0.9745 0.9906 0.3798 0.4395 0.3765 0.4977 0.4916 0.5578 0.4322 0.5006]

All p-values > 0.05. Cannot reject H_0 of no autocorrelation. Model is adequate.

Fitting GARCH(1,1) with Student-t errors for S&P 500...

S&P 500 fit complete. DoF (ν) = 5.03, AIC = 12221.01, BIC = 12253.01

Fitting GARCH(1,1) with Student-t errors for EURO STOXX 50...

EURO STOXX 50 fit complete. DoF (ν) = 5.37, AIC = 14053.80, BIC = 14085.80

Fitting GARCH(1,1) with Student-t errors for Nikkei 225...

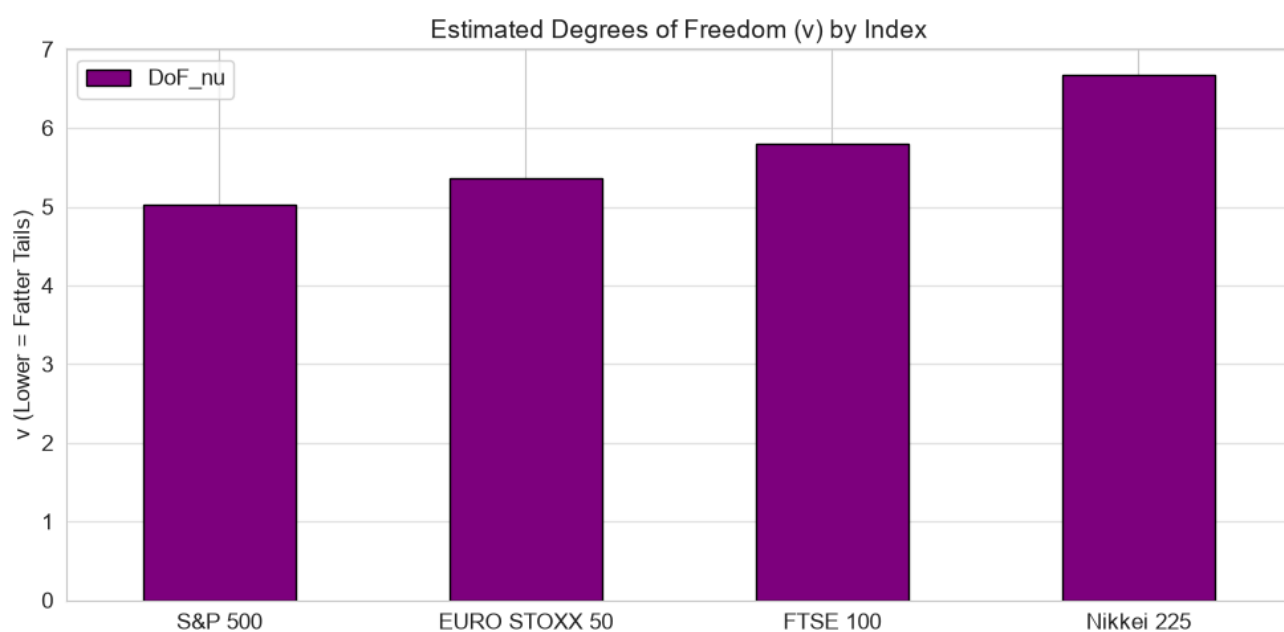
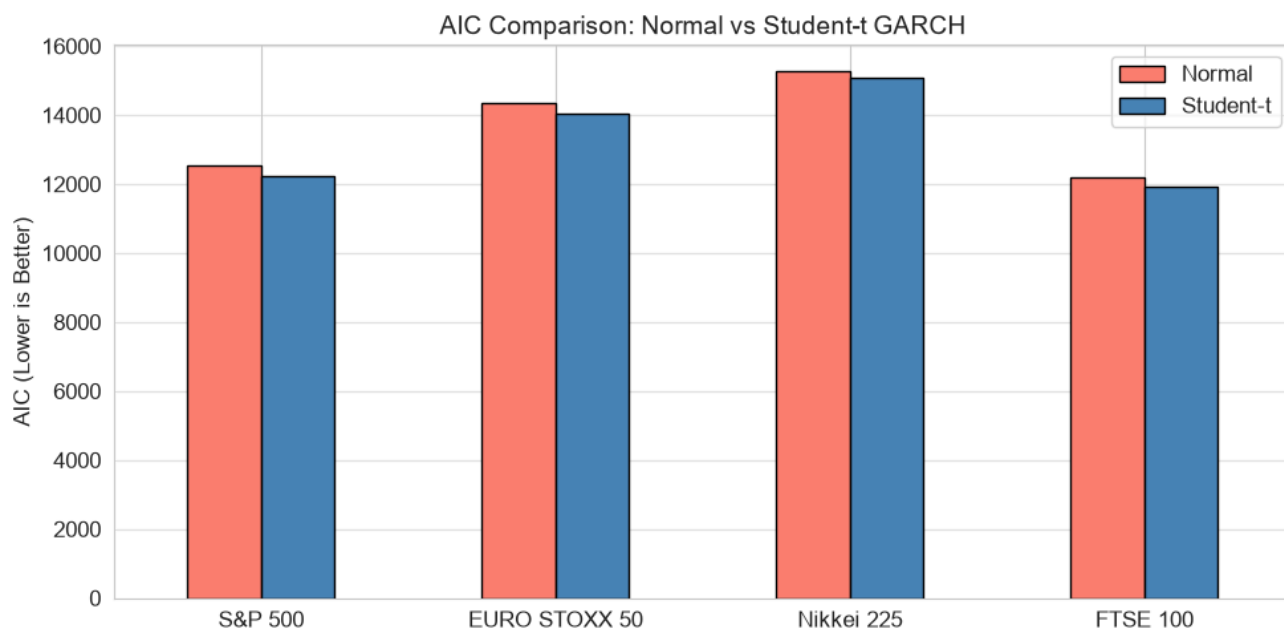
Nikkei 225 fit complete. DoF (ν) = 6.67, AIC = 15086.88, BIC = 15118.87

Fitting GARCH(1,1) with Student-t errors for FTSE 100...

FTSE 100 fit complete. DoF (ν) = 5.79, AIC = 11938.31, BIC = 11970.31

=== Normal vs Student-t Model Comparison ===

	AIC_Normal	AIC_t	DoF_nu	AIC_Improvement	Kurtosis_Implied
S&P 500	12555.45	12221.01	5.03	334.43	8.84
EURO STOXX 50	14344.71	14053.80	5.37	290.91	7.39
Nikkei 225	15265.62	15086.88	6.67	178.74	5.25
FTSE 100	12183.62	11938.31	5.79	245.31	6.35



=== 95% VaR Quantile Comparison (Normal vs Student-t) ===

S&P 500: $\nu = 5.03$ | Normal Quantile = -1.645 | t-Quantile = -2.013 | Risk Underestimated by 1.22x

EURO STOXX 50: $\nu = 5.37$ | Normal Quantile = -1.645 | t-Quantile = -1.985 | Risk Underestimated by 1.21x

Nikkei 225: $\nu = 6.67$ | Normal Quantile = -1.645 | t-Quantile = -1.909 | Risk Underestimated by 1.16x

FTSE 100: $\nu = 5.79$ | Normal Quantile = -1.645 | t-Quantile = -1.956 | Risk Underestimated by 1.19x